ASSIGNMENT 2

2.Take one Domain and draw the graph (Normal distribution) (Empirical Normal)

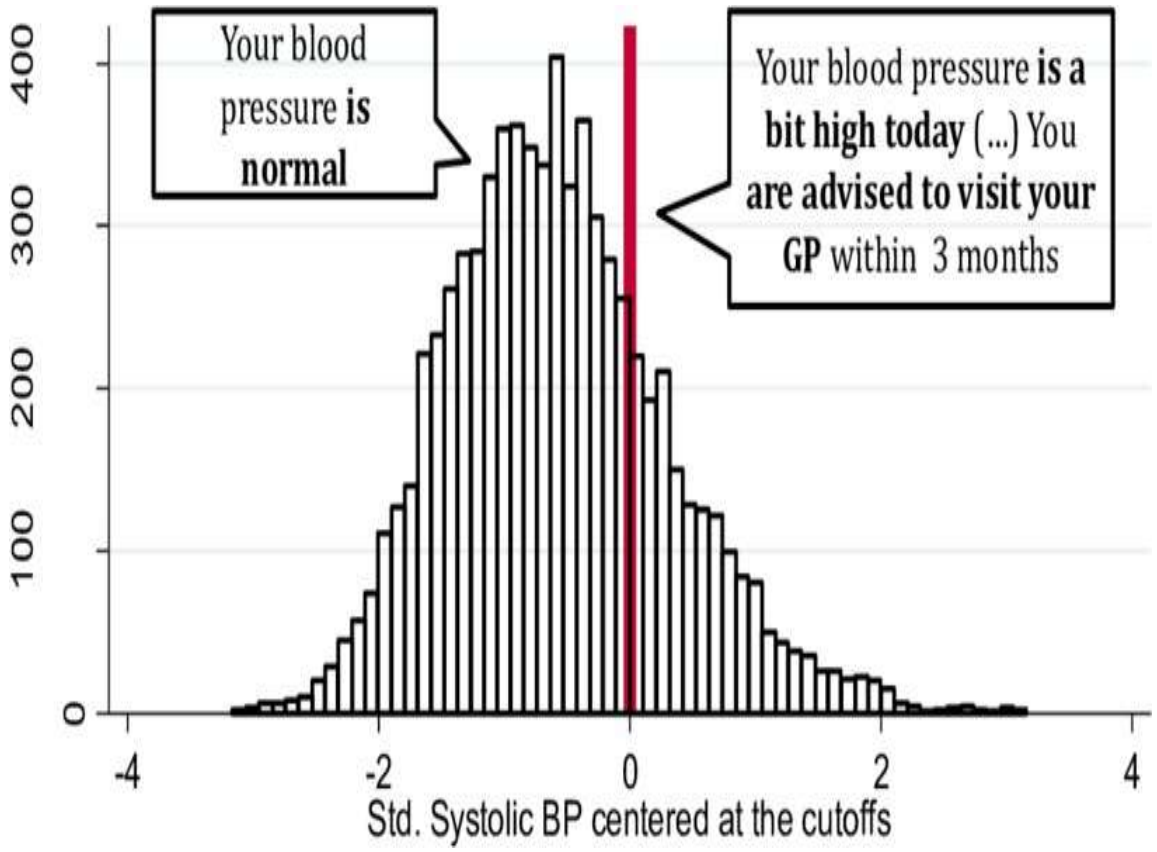
Domain: Distribution and Empirical Rule in the Healthcare

1. Introduction

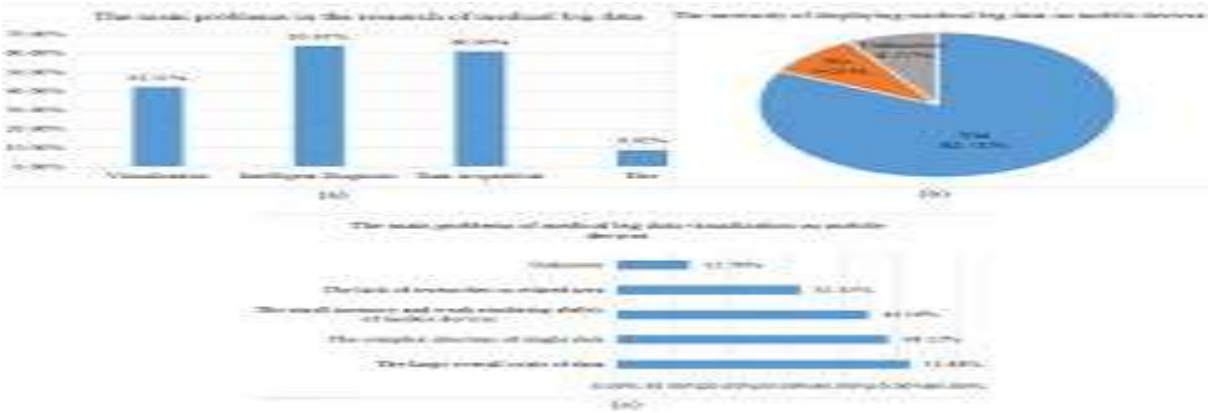
In healthcare, data analysis plays a crucial role in understanding patient health, diagnosing diseases, and planning treatments. Many biological and medical measurements such as **blood pressure, body temperature, cholesterol levels, and heart rate** follow a **Normal Distribution**.

Among these, **blood pressure readings** are one of the most commonly analyzed datasets. Most people have blood pressure values around a normal range, while very high or very low values occur less frequently. This natural pattern makes blood pressure an ideal example for explaining **Normal Distribution and the Empirical Rule**.

This report explains these statistical concepts using the **healthcare domain**, focusing on **blood pressure measurements**.



2. Domain Selection: Healthcare



Why Healthcare Domain?

Healthcare generates vast numerical data such as:

- Blood pressure readings
- Blood sugar levels
- Cholesterol levels
- Heart rate measurements

These datasets often follow normal distribution, helping doctors and researchers make reliable decisions.

3. Understanding Blood Pressure Data

What is Blood Pressure?

Blood pressure is the force exerted by circulating blood on the walls of blood vessels. It is measured as:

- **Systolic pressure** (upper value)
- **Diastolic pressure** (lower value)

Example: **120/80 mmHg**

In large populations, systolic blood pressure values typically form a normal distribution.

4. What is Normal Distribution?

Definition

A **Normal Distribution** is a probability distribution that:

- Is bell-shaped
- Is symmetric around the mean
- Has most values near the center

Key Characteristics

- Mean = Median = Mode
- Tails decrease equally on both sides
- Total probability = 1

5. Mathematical Model of Normal Distribution

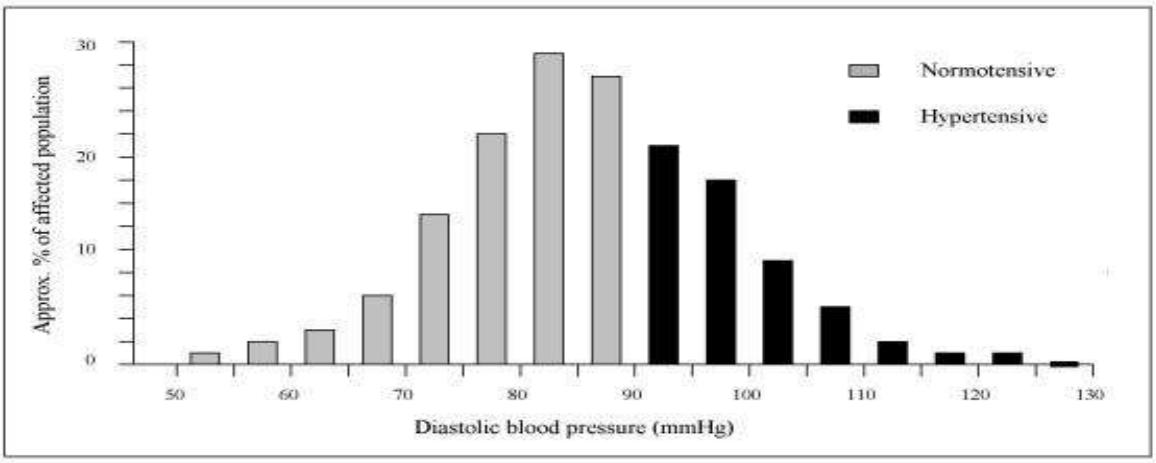
The normal distribution formula is:

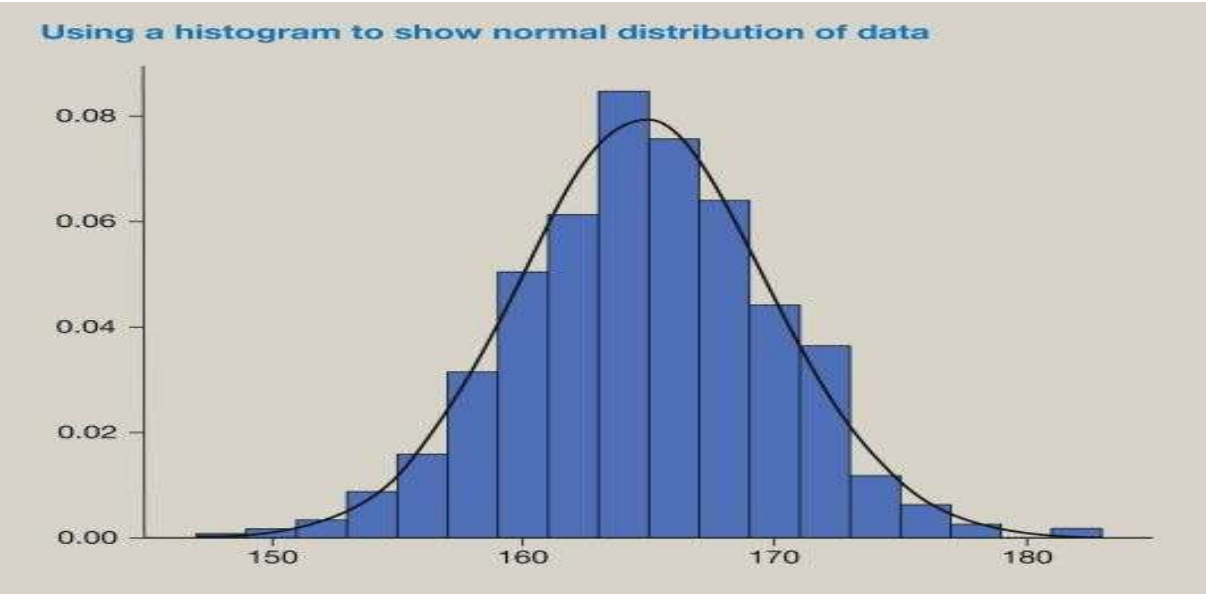
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

- μ = Mean blood pressure
- σ = Standard deviation
- x = Blood pressure value

6. Healthcare Example: Blood Pressure Distribution





Assumed Population Data

- Mean systolic BP (μ) = **120 mmHg**
- Standard deviation (σ) = **10 mmHg**

This represents a healthy adult population.

7. Drawing the Normal Distribution Graph (Explanation) X-Axis

- Blood pressure values (90–150 mmHg)

Y-A xis

- Number of patients / Probability density

Key Points

- $\mu - 3\sigma = 90$
- $\mu - 2\sigma = 100$
- $\mu - 1\sigma = 110$
- $\mu = 120$
- $\mu + 1\sigma = 130$
- $\mu + 2\sigma = 140$
- $\mu + 3\sigma = 150$

The curve peaks at 120 mmHg and gradually decreases on both sides.

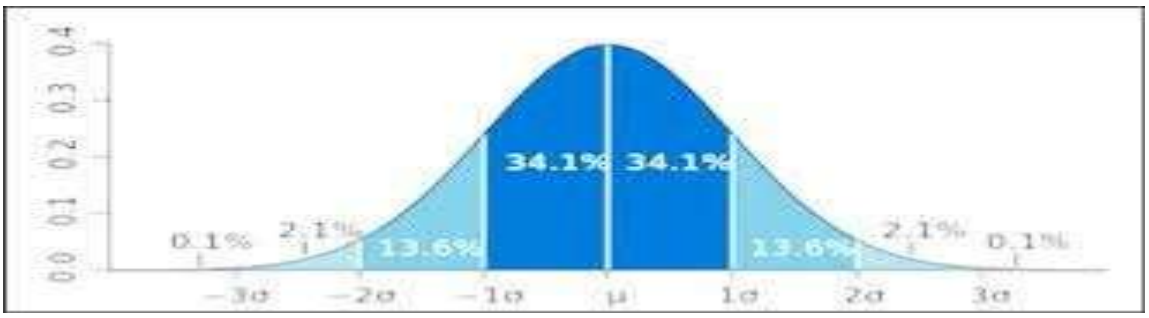
8. Meaning of Standard Deviation in Healthcare

Standard deviation indicates how much patients' blood pressure varies from the average.

- Low σ → Stable population health
- High σ → High variability, potential risk group

Doctors use this to assess population health trends.

G. Empirical Rule (68–G5–GG.7 Rule)



The empirical rule states:

- **68%** of values lie within **±1σ**
- **G5%** of values lie within **±2σ**
- **GG.7%** of values lie within **±3σ**

10. Applying Empirical Rule to Blood Pressure

Data 68% of Patients

$$120 \pm 10 = 110 \text{ to } 130$$

➞ 68% of patients have BP between **110 and 130 mmHg**

G5% of Patients

$$120 \pm 20 = 100 \text{ to } 140$$

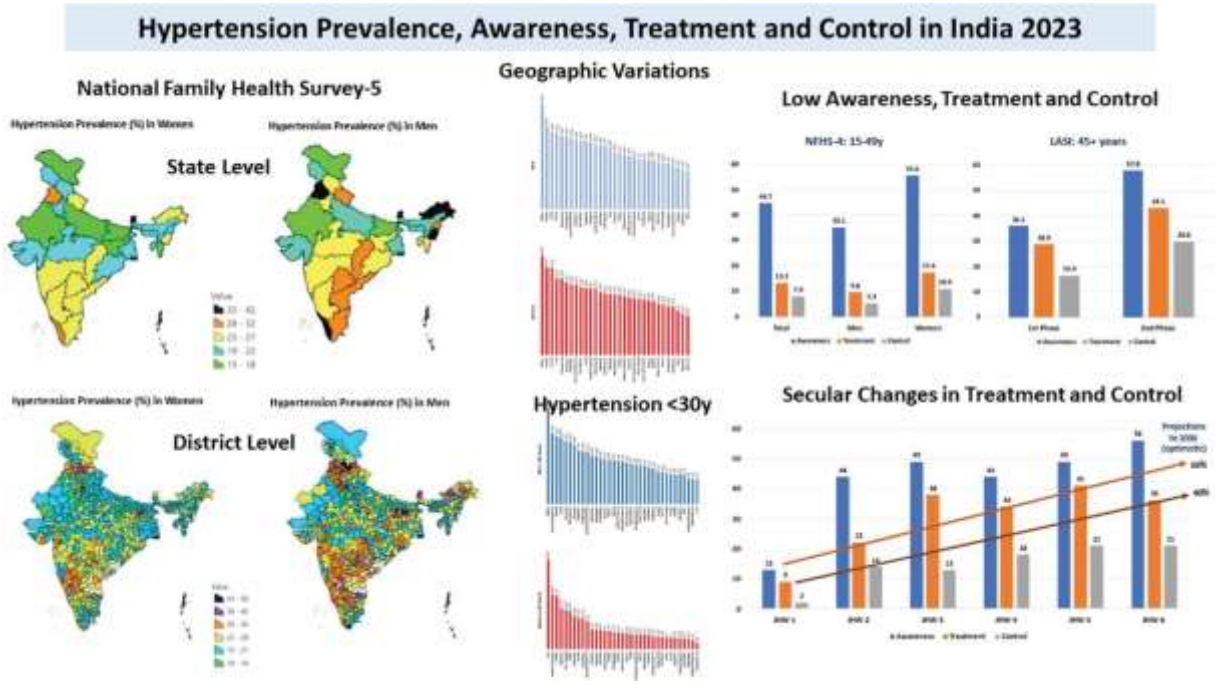
➞ 95% of patients fall between **100 and 140 mmHg**

GG.7% of Patients

$$120 \pm 30 = 90 \text{ to } 150$$

➞ Almost all patients fall in this range

11. Interpretation of Healthcare Results



- BP above **140 mmHg** → Hypertension risk
- BP below **G0 mmHg** → Hypotension risk
- Most patients lie in safe range

This helps in **early diagnosis and preventive care**.

12. Importance of Normal Distribution in Healthcare

- Disease risk identification
- Population health monitoring
- Clinical trial analysis
- Medical research validation
- Treatment effectiveness evaluation

13. How Doctors Use This in Practice

- Identify abnormal patients

- Decide medication dosage
- Design public health policies
- Predict future health trends

14. Advantages of Normal Distribution in Healthcare

- Easy visualization
- Reliable prediction
- Scientific validity
- Applicable to large populations

15. Limitations

- Individual cases may not follow normal pattern
- Outliers (critical patients) affect analysis
- Skewed data in small samples

16. Comparison with Skewed Medical Data

Aspect	Normal Data	Skewed Data
Shape	Bell curve	Left/Right skew
Empirical Rule	Works	Not reliable
Interpretation	Easy	Complex

17. Real-World Impact

- Hospitals allocate resources efficiently
- Governments plan health programs
- Researchers improve disease prevention

18.Summary

- Blood pressure follows normal distribution
- Mean and SD define population health
- Empirical rule simplifies diagnosis
- Extreme values indicate health risks

1G. Conclusion

The healthcare domain clearly demonstrates the importance of normal distribution and the empirical rule in analyzing medical data such as blood pressure. These statistical tools help doctors identify normal and abnormal health conditions, improve diagnosis accuracy, and support evidence-based medical decisions. Thus, normal distribution plays a vital role in modern healthcare analytics.